LEARNING MODULE

8

Vectors and Matrices

Holdings, prices and returns are vectors and a book of returns is a matrix: the dot product gives P&L and portfolio return, and the correlation of two return series is the cosine of the angle between their centred vectors.

VOLUME II · Linear Algebra for Portfolios

BUILDS ON LM1 · LM2 · LM3 · LM7

LEADS TO LM9 · LM10 · LM11 · LM12 · LM16 · LM23 · LM27 · LM28 · LM29 · LM31 · LM35

TIME ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 represent holdings, prices, weights and returns as vectors with a stated shape, unit per entry and index, and form linear combinations of them
☐	2 compute a book's P&L as $\mathbf{h}^\top \Delta\mathbf{P}$ and a portfolio's return as $\mathbf{w}^\top \mathbf{R}$, and explain why the second needs simple returns
☐	3 calculate norms, distances and unit vectors, and distinguish net exposure $\mathbf{1}^\top \mathbf{f}$ from gross exposure $\|\mathbf{f}\|_1$
☐	4 prove the Cauchy–Schwarz inequality with its equality case, and use it to define angles, orthogonality and the triangle inequality
☐	5 interpret a sample correlation as the cosine of the angle between centred return vectors, and bound the standard deviation of a long-only portfolio
☐	6 organize returns in a data matrix and compute a daily-rebalanced portfolio's returns $X\mathbf{w}$ and a fixed book's P&L $Y\mathbf{h}$
☐	7 multiply and transpose matrices, check their shapes, and apply associativity, non-commutativity and $(MK)^\top = K^\top M^\top$
☐	8 implement these operations in NumPy and Polars, with dates aligned by a join and assets in a fixed order

LEARNING MODULE OVERVIEW

- A vector is an ordered list with a unit for each entry and an index that says which asset, and which date, each entry belongs to. Two vectors add only when their orders, units and dates match.
- The dot product $\mathbf{x}^\top\mathbf{y} = \sum_k x_k y_k$ gives a book's P&L, $\Delta W = \mathbf{h}^\top \Delta\mathbf{P}$. On 7 April 2025 a book short one TX contract and long 0050 and 00679B made NT$57,400: the future and 0050 nearly cancelled, and the bond line made NT$54,000.
- Simple returns combine with weights, $R_p = \mathbf{w}^\top\mathbf{R}$, and log returns do not. The same day's P&L is exactly $W_0\mathbf{f}^\top\mathbf{R}$, 0.574% of equity, with net exposure $\mathbf{1}^\top\mathbf{f} = 0.295$ and gross exposure $\|\mathbf{f}\|_1 = 1.147$.
- The Cauchy–Schwarz inequality $|\mathbf{x}^\top\mathbf{y}| \le \|\mathbf{x}\|\|\mathbf{y}\|$ holds with equality only for parallel vectors, so every angle between non-zero vectors is well defined.
- A sample correlation is the cosine of the angle between centred return vectors: 13.0° for 0050 and 00631L, 96.1° for 0050 and 00679B, and 41.0° to 77.4° across the 378 pairs of the 28 leaf TWSE sector indices that FinMind serves daily.
- For long-only weights the triangle inequality gives $s_p \le \sum_i w_i s_i$: the 42.3/57.7 mix of 0050 and 00679B in LM7 had 10.91% annualized volatility against a weighted average of 16.29%.
- In a data matrix X, with a row per date and a column per asset, $X\mathbf{w}$ is a daily-rebalanced portfolio's return series and $Y\mathbf{h}$ a fixed book's daily P&L. Over 7–11 April the case book lost NT$124,200: $\mathbf{1}^\top Y\mathbf{h} = -124{,}200$.
- Matrix products follow a shape rule, are associative but not commutative, and transpose in reverse order. Summed daily returns are no period return: for a 60/40 mix of 0050 and 00679B the week's returns sum to −5.98% and compound to −6.32%.

1 INTRODUCTION

A trading book holds several instruments at once. Its holdings, prices, weights and returns are therefore lists, and its P&L and return are sums over them. Linear algebra writes such sums as products of vectors. It also adds geometry: a length that measures exposure and an angle that measures how alike two return series are. This module builds those tools on one real book and one real week.

MOTIVATING CASE **7 April 2025: two limit-downs and a bond that rose**

At the close of 2 April 2025 a trader with equity $W_0 = \text{NT\$10 million}$ holds the book $\mathbf{h} = (-1,\ 24{,}000,\ 100{,}000)^\top$. It is short one April TX contract and long 24,000 shares of 0050 and 100,000 shares of 00679B, the US Treasury ETF of LM7. The short future roughly offsets the 0050 line: NT$4.26 million of index exposure against NT$4.22 million.

Taiwan's markets were closed on 3 and 4 April and reopened on Monday 7 April, after the United States had announced new tariffs. TX and 0050 closed at their −10% limits, and 00679B rose (Exhibit 1).

How much did the book make on 7 April, and which line made it? Why did the "hedged" book still lose NT$124,200 over the week? How alike are the series, over a week and over a decade?

Exhibit 1: The case book on 2 and 7 April 2025

Line	Price unit	Holding h_i	2 April	7 April	Change (points or NT$)	Return	P&L $h_i \Delta P_i$ (NT$)
TX, April 2025 contract	index points, NT$200 a point	−1 contract	21,296	19,167	−2,129	−9.9972%	+425,800
0050	NT$ per share	24,000 shares	176.00	158.40	−17.60	−10.0000%	−422,400
00679B	NT$ per share	100,000 shares	29.88	30.42	+0.54	+1.8072%	+54,000
Book							**+57,400**

Note: TX at its daily settlement price, the ETFs at their closes. A TX contract gains or loses NT$200 a point, so its price change ΔP_i is $200 \times (-2{,}129) = -425{,}800$ NT$. The April contract's regular-session volume was 66,423 contracts on 2 April and 2,736 on 7 April. TAIEX fell 9.6997%, its largest daily move from July 2007 to October 2026. Source: FinMind `TaiwanFuturesDaily`, `TaiwanStockPrice` and `TaiwanStockDividendResult`, retrieved 10 October 2026 (TX, ETFs, TAIEX in 2025) and 11 October 2026 (TAIEX from July 2007); no ex-dividend date or split falls in the window. Computations: `data/lm08_numbers.py`.

Sections 2 and 3 introduce vectors and the dot product and answer the first question. Sections 4–6 add length and angle and show that a correlation is the cosine of an angle, which answers the third. Sections 7 and 8 stack the week into a matrix and answer the second.

NOTE **Notation for this module**

Vectors are bold columns and $^\top$ is the transpose. Local symbols: holdings $\mathbf{h}$ (contracts, shares); prices $\mathbf{P}_t$ in NT$ per unit, TX at $200F_t$ for a settlement price of F_t points, and $\Delta\mathbf{P}_t := \mathbf{P}_t - \mathbf{P}_{t-1}$; equity W_0 and the P&L ΔW_t of date t; log returns $\mathbf{r}$; generic vectors $\mathbf{x}$, $\mathbf{y}$, a unit vector $\mathbf{u}$, a generic number a and the best one, a^*, in a proof; the 1-norm $\|\mathbf{x}\|_1$. For T returns: the mean $\bar{x}$ of the entries, the centred vector $\tilde{\mathbf{x}} := \mathbf{x} - \bar{x}\mathbf{1}$, standard deviations s_x, s_i (asset i) and s_p (a portfolio) with divisor T as in LM3, and the sample correlation ρ. The $T \times N$ return matrix X has entries x_{tj}, rows $\mathbf{R}_t^\top$ and columns $\mathbf{x}_j$; Y has rows $\Delta\mathbf{P}_t^\top$; M and K are generic matrices with entries m_{ij} and k_{ij}. Counts: N assets, T returns (never a horizon), n rows, m portfolios; i, j assets, rows or columns, t dates, k entries and ℓ the inner index of a product.

2 VECTORS: HOLDINGS, PRICES, WEIGHTS AND RETURNS

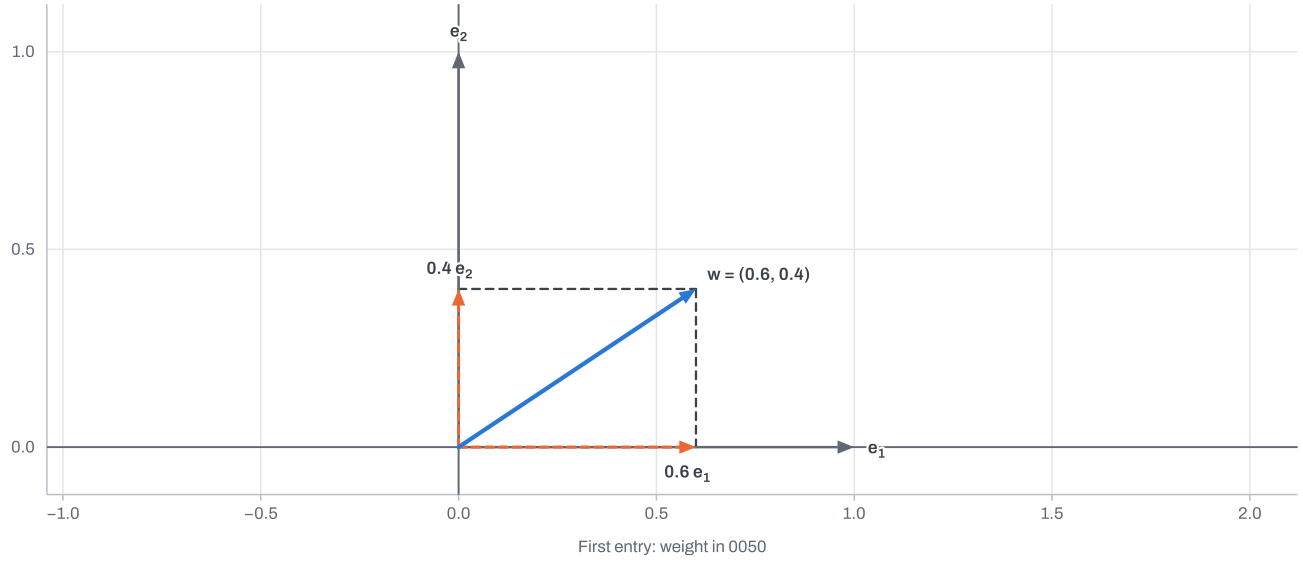
- [ ] represent holdings, prices, weights and returns as vectors with a stated shape, unit per entry and index, and form linear combinations of them

A **vector** is an ordered list of numbers, written as a column $\mathbf{x} = (x_1, \dots, x_N)^\top$. Entry k always refers to the same thing, here the same asset, so the order is part of the vector. Exhibit 2 draws the weights $\mathbf{w} = (0.6, 0.4)^\top$ of a mix of 0050 and 00679B as an arrow from the origin to the point $(0.6, 0.4)$. The arrow is a sum of two scaled basis vectors:

$$\begin{aligned}\mathbf{w} = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} &= 0.6 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0.4 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= 0.6\,\mathbf{e}_1 + 0.4\,\mathbf{e}_2.\end{aligned}$$

Exhibit 2: A weight vector as an arrow and as a list

Second entry: weight in 00679B



Note: $\mathbf{w} = (0.6, 0.4)^{\top}$ holds 60% in 0050 and 40% in 00679B. The dashed arrows are its components $0.6\,\mathbf{e}_1$ and $0.4\,\mathbf{e}_2$, and the grey arrows the basis vectors.

Shape, unit and index. A vector from a market needs three labels besides its numbers. Its shape is the number of entries. Each entry has a unit, and an index says which asset each entry belongs to and, for prices and returns, on which date. In the case every vector lists (TX, 0050, 00679B) in that order:

$$\mathbf{h} = \begin{pmatrix} -1 \\ 24{,}000 \\ 100{,}000 \end{pmatrix},$$

$$\mathbf{P}_{2\ \mathrm{Apr}} = \begin{pmatrix} 200 \times 21{,}296 \\ 176.00 \\ 29.88 \end{pmatrix},$$

$$\mathbf{R}_{7\ \mathrm{Apr}} = \begin{pmatrix} -0.0999718 \\ -0.1000000 \\ 0.0180723 \end{pmatrix}.$$

Addition and multiples. Vectors of the same shape add entry by entry, and a number a multiplies every entry: $(\mathbf{x} + \mathbf{y})_k = x_k + y_k$ and $(a\mathbf{x})_k = a x_k$. A sum such as $w_1\mathbf{x}_1 + \cdots + w_N\mathbf{x}_N$, with numbers w_j and vectors $\mathbf{x}_j$, is a **linear combination**. Holding two books together adds their holdings, and Section 7 builds every portfolio's return series in the same way.

Two vectors can be added only when they agree in asset order, in the unit of each entry and, for prices and returns, in date. The sum $\mathbf{h} + \mathbf{P}_{2\ \mathrm{Apr}}$ has the right shape but no meaning.

EXAMPLE 1

Shapes, units and index of the case's vectors

State the shape, the unit of each entry and the index of $\mathbf{h}$, $\mathbf{P}_t$, the exposures $\mathbf{f}$ (position values ÷ equity, LM2) and $\mathbf{R}_t$ in the case. A second trader holds $(-1,\ 20{,}000,\ 100{,}000)^\top$. Write the desk's combined book, and the difference between the two books as a multiple of a basis vector.

SOLUTION

All four have three entries in the order (TX, 0050, 00679B). The holdings count contracts, shares and shares, and the prices are NT$ per contract ($200F_t$) and per share at the close of date t. The exposures and returns are pure numbers: $\mathbf{f}$ is measured at the close of 2 April, and $\mathbf{R}_t$ covers the session that ends on date t.

The combined book is $(-2,\ 44{,}000,\ 200{,}000)^\top$. The difference $\mathbf{h} - (-1,\ 20{,}000,\ 100{,}000)^\top$ is $4{,}000\,\mathbf{e}_2$: the books differ only in 0050.

PITFALL **Two lists in different orders or on different dates**

Two columns of closes can look alike and still be misaligned. 0050 was halted from 11 to 17 June 2025 for its split while 00679B traded. Stacked by position, the two files would pair returns of different days. Join on the date first: an inner join keeps the 2,364 dates that both funds share from 17 January 2017, which give the 2,363 returns of LM7. Fix the asset order once and keep it in every vector.

KNOWLEDGE CHECK 1

A colleague sends holdings in the order (00679B, 0050, TX) as $(100{,}000,\ 24{,}000,\ -1)^\top$. Can you add this vector to $\mathbf{h}$? What is the sum?

ANSWER

Not as sent: its first entry counts 00679B shares, not TX contracts. Reordered to (TX, 0050, 00679B), it equals $\mathbf{h}$, so the sum is $2\mathbf{h} = (-2,\ 48{,}000,\ 200{,}000)^\top$.

3 THE DOT PRODUCT: P&L AND PORTFOLIO RETURN

- ☐ compute a book's P&L as $\mathbf{h}^\top \Delta\mathbf{P}$ and a portfolio's return as $\mathbf{w}^\top \mathbf{R}$, and explain why the second needs simple returns

Multiply two vectors entry by entry and add the products: the result is one number.

DEFINITION **Dot product**

The **dot product** of two vectors of the same shape is

$$\mathbf{x}^\top \mathbf{y} := \sum_{k=1}^{N} x_k y_k = x_1 y_1 + \cdots + x_N y_N. \tag{8.1}$$

The dot product is also written $\mathbf{x} \cdot \mathbf{y}$. The form $\mathbf{x}^\top \mathbf{y}$ reads as a row times a column, a matrix product (Section 8).

P&L. A book's P&L on date t adds, over its lines, the holding times the price change:

$$\Delta W_t = \sum_{i=1}^{N} h_i \, \Delta P_{i,t} = \mathbf{h}^\top \Delta \mathbf{P}_t. \tag{8.2}$$

Each term is contracts times NT$ per contract, or shares times NT$ per share, so every term and the sum are in NT$. The units must match line by line, which is why TX enters at $200 F_t$ NT$ per contract. Interest on cash and on margin deposits is ignored.

EXAMPLE 2

The book's P&L on 7 April 2025, line by line

Use Exhibit 1 to compute $\mathbf{h}^\top \Delta \mathbf{P}$ for 7 April, and say which line made the money.

SOLUTION

The price changes are $200 \times (19{,}167 - 21{,}296) = -425{,}800$ NT$ per TX contract, $158.40 - 176.00 = -17.60$ and $30.42 - 29.88 = 0.54$ NT$ per share. So

$$\begin{aligned} \mathbf{h}^\top \Delta \mathbf{P} &= (-1)(-425{,}800) \\ &\quad + 24{,}000\,(-17.60) \\ &\quad + 100{,}000\,(0.54) \\ &= 425{,}800 - 422{,}400 + 54{,}000 \\ &= 57{,}400. \end{aligned}$$

The book made NT$57,400. The short future and the 0050 line nearly cancel, leaving NT$3,400, so the bond line made almost all of the gain. The hedge is sized by notional value; LM23 sizes hedges by regression, with the least-squares tools of LM12.

Portfolio return. LM1 (1.13) showed that simple returns add across assets with weights. Take a fully invested book, so that $W_0 = \sum_i h_i P_{i,0}$, with weights $w_i := h_i P_{i,0} / W_0$. Then

$$1 + R_p = \frac{\sum_i h_i P_{i,1}}{\sum_i h_i P_{i,0}} = \sum_i \frac{h_i P_{i,0}}{W_0}\,\frac{P_{i,1}}{P_{i,0}}$$
$$= \sum_i w_i (1 + R_i) = 1 + \mathbf{w}^\top \mathbf{R},$$

where the last step uses $\sum_i w_i = 1$. Hence

$$R_p = \mathbf{w}^\top \mathbf{R}. \tag{8.3}$$

Log returns do not combine this way: a portfolio's log return is $\ln(1 + \mathbf{w}^\top \mathbf{R})$, not $\mathbf{w}^\top \mathbf{r}$ (LM1 (1.14)).

Exposures. A book with futures breaks the identity $W_0 = \sum_i h_i P_{i,0}$, because a futures contract costs nothing to enter. Define each line's exposure as its position value over equity, $f_i := h_i P_{i,0} / W_0$, the f of LM2. Line by line, $h_i \Delta P_i = W_0 f_i R_i$, because

$$f_i R_i = \frac{h_i P_{i,0}}{W_0} \cdot \frac{\Delta P_i}{P_{i,0}} = \frac{h_i \Delta P_i}{W_0}.$$

Summing over the lines gives

$$\Delta W = \mathbf{h}^\top \Delta \mathbf{P} = W_0\, \mathbf{f}^\top \mathbf{R}, \tag{8.4}$$

exactly, for one period and whatever the net exposure $\mathbf{1}^\top \mathbf{f}$ is. For a fully invested book $\mathbf{f} = \mathbf{w}$, and (8.4) is (8.3) times W_0.

EXAMPLE 3

The same P&L from exposures and returns

Compute the exposures $\mathbf{f}$ at the close of 2 April, and check (8.4) for 7 April.

SOLUTION

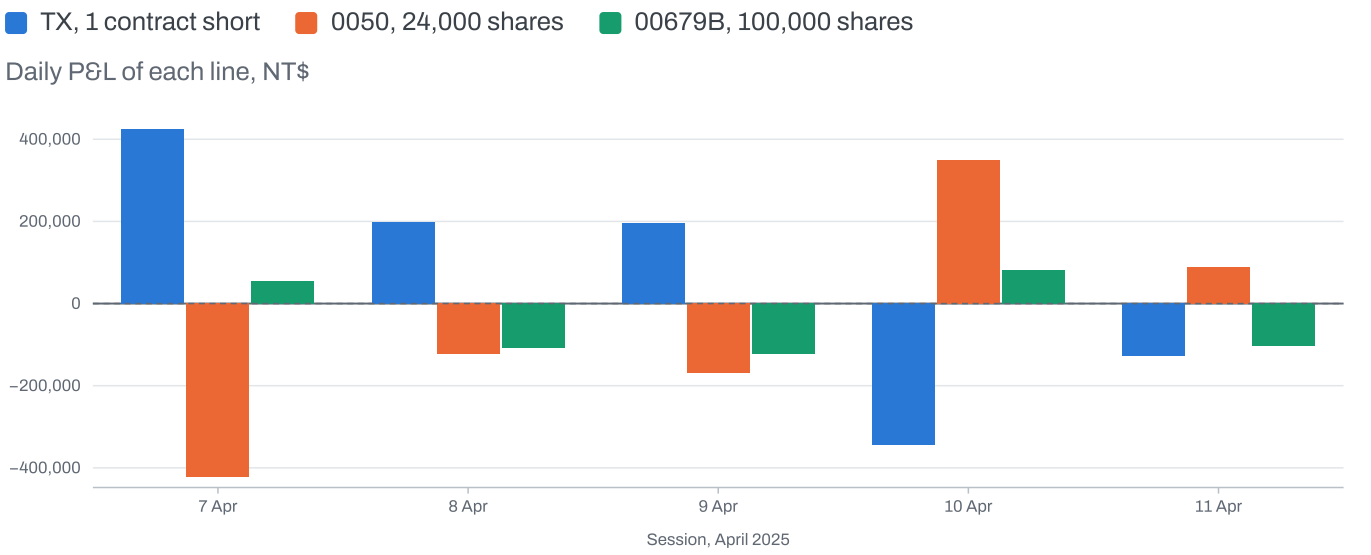
The position values $h_i P_{i,0}$ are $-4{,}259{,}200$, $4{,}224{,}000$ and $2{,}988{,}000$ NT\$, so $\mathbf{f} = (-0.42592,\ 0.4224,\ 0.2988)^\top$. With the returns of Exhibit 1, the products $f_i R_i$ are 0.042580, -0.042240 and 0.005400. They sum to $\mathbf{f}^\top \mathbf{R} = 0.005740$, a return of 0.574% on equity, and $W_0\, \mathbf{f}^\top \mathbf{R} =$ NT\$57,400. No rounding is involved: each product $f_i R_i$ equals $h_i \Delta P_i / W_0$ exactly.

Two rules make the dot product easy to use. It is symmetric, $\mathbf{x}^\top \mathbf{y} = \mathbf{y}^\top \mathbf{x}$, and it is linear in each vector:

$$(a\mathbf{x} + \mathbf{y})^\top \Delta \mathbf{P} = a\, \mathbf{x}^\top \Delta \mathbf{P} + \mathbf{y}^\top \Delta \mathbf{P}.$$

So two books held together make the sum of their P&Ls. The P&L over several days is likewise the sum of the daily P&Ls, which Exhibit 3 shows for the week.

Exhibit 3: Daily P&L of each line of the case book, 7–11 April 2025



Note: $h_i \Delta P_{i,t}$ in NT$ for the fixed book $\mathbf{h} = (-1,\ 24{,}000,\ 100{,}000)^\top$; TX at NT$200 a point. The book's daily totals are +57,400, −33,000, −95,200, +87,800 and −141,200. Source: FinMind `TaiwanFuturesDaily` and `TaiwanStockPrice`, retrieved 10 October 2026.

PITFALL **A limit-locked P&L is a mark, not cash**

TX may move at most 10% from the previous regular session's settlement price, in ticks of one index point (TAIFEX). On 7 April that limit, $0.9 \times 21{,}296 = 19{,}166.4$, fell between two ticks. TX traded and settled at 19,167, a rounding that the data show and TAIFEX's pages do not state. The settlement price is the volume-weighted price of the last minute's trades, and TAIFEX states no special rule for a locked market.

The April contract's regular-session volume fell from 66,423 contracts on 2 April to 2,736, so the settlement of 19,167 came from thin trading. On 10 April, when TX settled at its +10% limit, the volume was 2,382.

In a market locked at its lower limit, sell orders queue while a buy order fills at once. A buy order would have closed the short TX line at 19,167 and realized its +425,800. A sell order for the long 0050 line, however, would only have joined the queue at 158.40, TWSE's 10% limit. The book could not be closed at its marks, so treat a limit day's P&L as a valuation, not as cash.

導讀 **內積就是損益**

內積（dot product）把兩個向量（vector）逐項相乘再加總。持有部位乘上價格變動，加總就是整本帳的損益 $\Delta W = \mathbf{h}^\top \Delta \mathbf{P}$。權重乘上簡單報酬（simple return），加總就是投資組合報酬；對數報酬（log return）不能這樣加權相加。4 月 7 日這本帳賺了新臺幣 57,400 元：期貨空單與 0050 幾乎互相抵銷，賺錢的是美債 ETF。跌停鎖死時，賣單排隊、難以成交，買單卻能立刻成交：期貨空單可以在 19,167 回補、實現獲利，0050 多單的賣單卻只能在 158.40 排隊，所以整本帳的損益只是評價。

KNOWLEDGE CHECK 2

A second trader holds $(-1,\ 20{,}000,\ 100{,}000)^\top$ from the close of 2 April. Compute this book's P&L on 7 April line by line. Why does it make more than the case book?

ANSWER

The lines are $+425{,}800$, $20{,}000 \times (-17.60) = -352{,}000$ and $+54{,}000$, so the P&L is NT$127,800. It holds 4,000 fewer shares of 0050, which fell NT$17.60 a share, so it loses NT$70,400 less on that line.

4 LENGTH, DISTANCE AND EXPOSURE

- [ ] calculate norms, distances and unit vectors, and distinguish net exposure $\mathbf{1}^\top \mathbf{f}$ from gross exposure $\|\mathbf{f}\|_1$

The dot product of a vector with itself is a sum of squares, and its square root is the vector's length.

DEFINITION **Norm, distance and unit vector**

The **Euclidean norm** (length) of $\mathbf{x}$ is

$$\|\mathbf{x}\| := \sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{x_1^2 + \cdots + x_N^2}, \tag{8.5}$$

and the **distance** between $\mathbf{x}$ and $\mathbf{y}$ is $\|\mathbf{x} - \mathbf{y}\|$. A **unit vector** has length 1, and for $\mathbf{x} \neq \mathbf{0}$ the unit vector in the direction of $\mathbf{x}$ is $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$. The **1-norm** is $\|\mathbf{x}\|_1 := \sum_k |x_k|$.

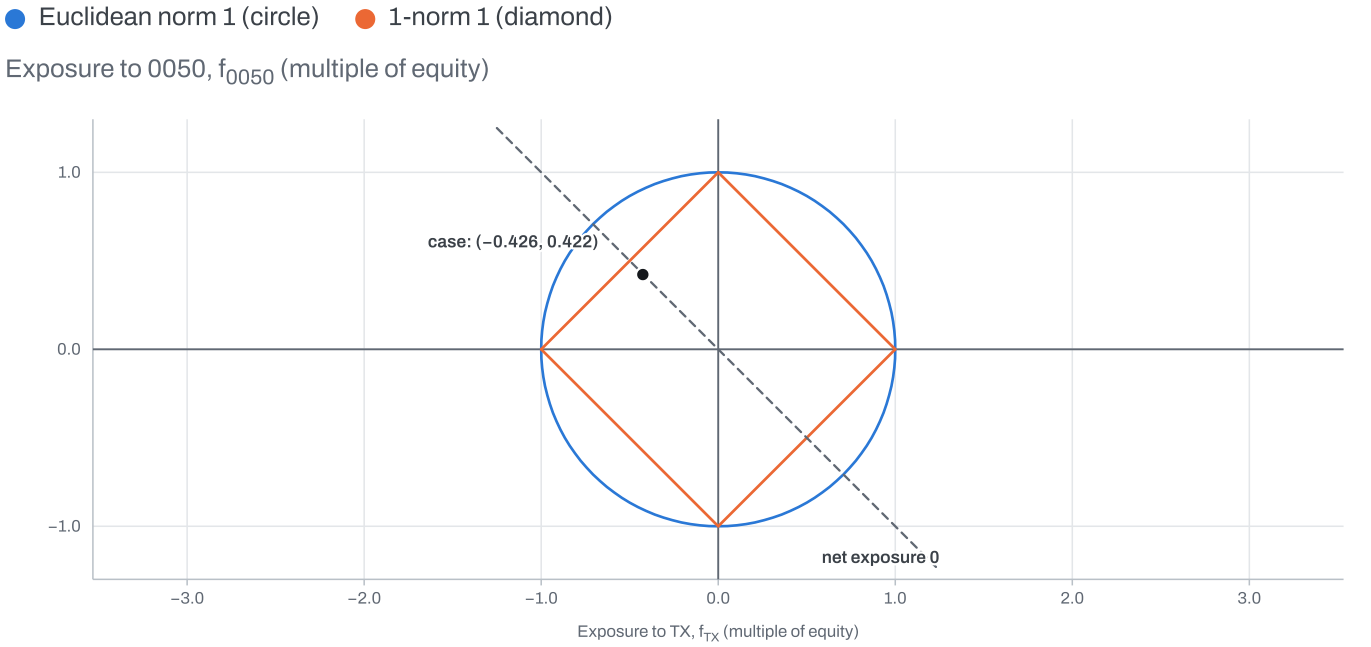
In two dimensions (8.5) is Pythagoras' theorem: the arrow of Exhibit 2 has length $\sqrt{0.6^2 + 0.4^2} = 0.7211$. A norm scales with its vector, $\|a\mathbf{x}\| = |a|\,\|\mathbf{x}\|$, and it is zero only for $\mathbf{x} = \mathbf{0}$.

Net and gross exposure. Two numbers summarize a book's exposures $\mathbf{f}$:

$$\text{net exposure} = \mathbf{1}^\top \mathbf{f} = \sum_i f_i,$$
$$\text{gross exposure} = \|\mathbf{f}\|_1 = \sum_i |f_i|.$$

Net exposure lets a short cancel a long, while gross exposure adds both. The two agree only when every position is long. Exhibit 4 draws both norms for the case's TX and 0050 exposures.

Exhibit 4: Two norms and the case's equity exposures



Note: the circle holds the points of Euclidean norm 1 and the diamond the points of 1-norm 1. The dot is the case's $(f_{\text{TX}}, f_{0050}) = (-0.426,\ 0.422)$: Euclidean norm 0.600, 1-norm 0.848 and net exposure −0.004, almost on the dashed line of zero net exposure.

EXAMPLE 4

Net and gross exposure, and the distance between two mixes

(a) Compute the net and gross exposure of $\mathbf{f} = (-0.42592,\ 0.4224,\ 0.2988)^\top$. (b) How far apart are the weight vectors $(0.6, 0.4)^\top$ and $(0.5, 0.5)^\top$?

SOLUTION

(a) Net: $-0.42592 + 0.4224 + 0.2988 = 0.29528$. Gross: $0.42592 + 0.4224 + 0.2988 = 1.14712$. On equity of NT$10 million, the book has NT$2,952,800 of net and NT$11,471,200 of gross exposure. The hedge cancels in the net figure but not in the gross one.

(b) The difference is $(0.1, -0.1)^\top$, so the distance is $\sqrt{0.1^2 + 0.1^2} = 0.1414$. Moving 10 points of weight from 00679B to 0050 is a step of length 0.1414.

PITFALL **Net exposure is not gross exposure**

A limit on net exposure lets a short offset more longs, while a limit on gross exposure counts the short as more exposure. The case book's net 0.295 hides a gross 1.147. Before comparing a book with a limit, check which of the two the limit names; LM7 drew the same distinction for a leverage cap.

KNOWLEDGE CHECK 3

A long–short pair has exposures $\mathbf{f} = (0.5,\ -0.5)^\top$. Find its net and gross exposure, its Euclidean norm and the unit vector in its direction.

ANSWER

Net $0.5 - 0.5 = 0$, gross $0.5 + 0.5 = 1$ and norm $\sqrt{0.25 + 0.25} = 0.7071$. The unit vector is $\mathbf{f}/0.7071 = (0.7071,\ -0.7071)^\top$.

5 ANGLES, THE CAUCHY–SCHWARZ INEQUALITY AND ORTHOGONALITY

- ☐ prove the Cauchy–Schwarz inequality with its equality case, and use it to define angles, orthogonality and the triangle inequality

In the plane, the law of cosines links the sides of the triangle formed by $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ to the angle θ between $\mathbf{x}$ and $\mathbf{y}$. Expanding the squared length of $\mathbf{x} - \mathbf{y}$ with the dot product gives the same formula, with $\mathbf{x}^\top \mathbf{y}$ in place of $\|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$:

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\|^2 &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\,\mathbf{x}^\top \mathbf{y} \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\|\mathbf{y}\| \cos\theta. \end{aligned}$$

So in the plane $\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$. In N dimensions this equation becomes the definition of the angle, once we know that the ratio of $\mathbf{x}^\top \mathbf{y}$ to $\|\mathbf{x}\|\|\mathbf{y}\|$ lies between −1 and 1.

THEOREM **Cauchy–Schwarz inequality**

For any two vectors $\mathbf{x}$ and $\mathbf{y}$ of the same shape,

$$|\mathbf{x}^\top \mathbf{y}| \le \|\mathbf{x}\|\ \|\mathbf{y}\|, \tag{8.6}$$

with equality exactly when one of the two vectors is a scalar multiple of the other (Strang 2016).

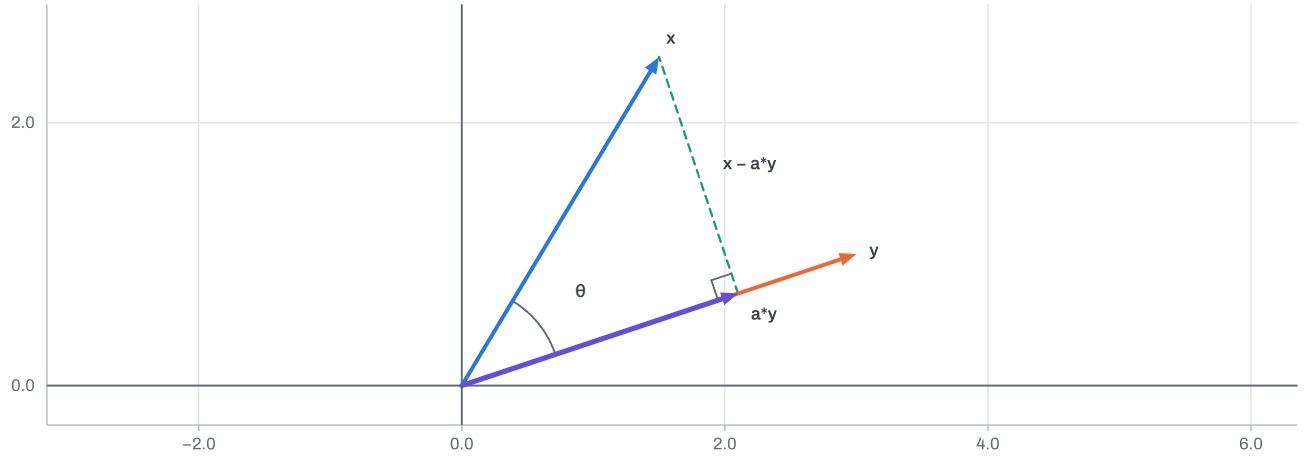
Proof. If $\mathbf{y} = \mathbf{0}$, both sides are 0, and $\mathbf{y} = 0 \cdot \mathbf{x}$. Otherwise the squared length of $\mathbf{x} - a\mathbf{y}$ is non-negative for every number a, and the choice $a^* := \mathbf{x}^\top \mathbf{y} / \|\mathbf{y}\|^2$ gives

$$\begin{aligned} 0 \le \|\mathbf{x} - a^*\mathbf{y}\|^2 &= \|\mathbf{x}\|^2 - 2a^*\,\mathbf{x}^\top\mathbf{y} + a^{*2}\|\mathbf{y}\|^2 \\ &= \|\mathbf{x}\|^2 - \frac{(\mathbf{x}^\top\mathbf{y})^2}{\|\mathbf{y}\|^2}. \end{aligned}$$

Multiplying by $\|\mathbf{y}\|^2$ gives $(\mathbf{x}^\top\mathbf{y})^2 \le \|\mathbf{x}\|^2\|\mathbf{y}\|^2$, and square roots give (8.6). Equality holds exactly when $\|\mathbf{x} - a^*\mathbf{y}\| = 0$, that is, when $\mathbf{x} = a^*\mathbf{y}$. Conversely, if $\mathbf{x} = a\mathbf{y}$ for some number a, both sides of (8.6) equal $|a|\,\|\mathbf{y}\|^2$. ■

The quadratic $\|\mathbf{x}\|^2 - 2a\,\mathbf{x}^\top\mathbf{y} + a^2\|\mathbf{y}\|^2$ is smallest where its derivative in a is zero, at a^*. So $a^*\mathbf{y}$ is the multiple of $\mathbf{y}$ closest to $\mathbf{x}$, the projection of $\mathbf{x}$ on $\mathbf{y}$ (LM12). Exhibit 5 draws it: the leftover $\mathbf{x} - a^*\mathbf{y}$ is perpendicular to $\mathbf{y}$.

Exhibit 5: The proof in a picture



Note: $\mathbf{x} = (1.5,\ 2.5)^\top$ and $\mathbf{y} = (3,\ 1)^\top$, so $a^* = \mathbf{x}^\top\mathbf{y}/\|\mathbf{y}\|^2 = 7/10$ and $a^*\mathbf{y} = (2.1,\ 0.7)^\top$. The dashed leftover $\mathbf{x} - a^*\mathbf{y} = (-0.6,\ 1.8)^\top$ is perpendicular to $\mathbf{y}$, and $\theta = 40.60°$.

With entries indexed by dates, (8.6) is the sum form that LM7 stated without proof:

$$\left|\sum_t x_t y_t\right| \le \sqrt{\sum_t x_t^2}\,\sqrt{\sum_t y_t^2}.$$

DEFINITION **Angle and orthogonality**

For non-zero vectors $\mathbf{x}$ and $\mathbf{y}$, the **angle** θ between them, from 0° to 180°, is defined by

$$\cos\theta := \frac{\mathbf{x}^\top\mathbf{y}}{\|\mathbf{x}\|\,\|\mathbf{y}\|}. \tag{8.7}$$

Two vectors are **orthogonal**, $\mathbf{x} \perp \mathbf{y}$, when $\mathbf{x}^\top\mathbf{y} = 0$.

By (8.6) the ratio in (8.7) lies between −1 and 1, so (8.7) defines an angle for every pair of non-zero vectors. The angle is 0° when $\mathbf{y}$ is a positive multiple of $\mathbf{x}$, 180° for a negative multiple and 90° for orthogonal vectors. A **dollar-neutral** book, with net exposure $\mathbf{1}^\top \mathbf{f} = 0$, is orthogonal to $\mathbf{1}$.

Triangle inequality. The length of a sum is at most the sum of the lengths. Expand the square and apply (8.6) to the middle term:

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\|^2 &= \|\mathbf{x}\|^2 + 2\,\mathbf{x}^\top\mathbf{y} + \|\mathbf{y}\|^2 \\ &\le \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= \big(\|\mathbf{x}\| + \|\mathbf{y}\|\big)^2, \end{aligned}$$

so that

$$\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|. \tag{8.8}$$

EXAMPLE 5

Two angles: two weight mixes, and a hedge against the vector of ones

(a) Find the angle between $\mathbf{x} = (0.6, 0.4)^\top$ and $\mathbf{y} = (0.5, 0.5)^\top$, and check (8.6). (b) The case's equity lines have exposures $(f_{\mathrm{TX}}, f_{0050}) = (-0.42592,\ 0.4224)$. Find their angle with $\mathbf{1} = (1, 1)^\top$.

SOLUTION

(a) The dot product is $0.30 + 0.20 = 0.50$, and $\|\mathbf{x}\|\|\mathbf{y}\| = 0.72111 \times 0.70711 = 0.50990$, so (8.6) holds. Then $\cos\theta = 0.50/0.50990 = 0.98058$ and $\theta = 11.31°$.

(b) The dot product with $\mathbf{1}$ is the net exposure, $-0.42592 + 0.4224 = -0.00352$. The lengths are 0.59986 and $\sqrt{2} = 1.41421$, so $\cos\theta = -0.00352/0.84833 = -0.00415$ and $\theta = 90.24°$. The pair is almost orthogonal to $\mathbf{1}$, because the short future nearly cancels the 0050 line. An exactly dollar-neutral pair such as $(-0.4224,\ 0.4224)$ makes exactly 90°.

導讀 **夾角與 Cauchy–Schwarz 不等式**

在平面上，兩個箭頭的內積（dot product）等於兩個長度相乘、再乘上夾角（angle）的餘弦（cosine）。Cauchy–Schwarz 不等式（Cauchy–Schwarz inequality）保證內積的絕對值不超過兩個長度的乘積，所以在任何維度都能用 $\cos\theta = \mathbf{x}^\top\mathbf{y}/(\|\mathbf{x}\|\|\mathbf{y}\|)$ 定義夾角。證明只用一件事：$\mathbf{x} - a\mathbf{y}$ 的長度平方不會是負的，挑最好的 a 就得到不等式；等號只在其中一個向量（vector）是另一個的倍數時成立。內積為零就是正交（orthogonal），淨曝險為零的多空部位與全為 1 的向量正交。三角不等式（triangle inequality）也是它的直接結果：和的長度不超過長度的和。

KNOWLEDGE CHECK 4

For $\mathbf{x} = (0.6, 0.4)^\top$ and $\mathbf{y} = (0.5, 0.5)^\top$, find the number a for which $\mathbf{x} - a\mathbf{y}$ is orthogonal to $\mathbf{y}$. Interpret $\mathbf{x} - a\mathbf{y}$.

ANSWER

The proof's choice works: $a^* = \mathbf{x}^\top\mathbf{y}/\|\mathbf{y}\|^2 = 0.50/0.50 = 1$. Then $\mathbf{x} - \mathbf{y} = (0.1,\ -0.1)^\top$, and $(0.1)(0.5) + (-0.1)(0.5) = 0$. The 60/40 mix is the 50/50 mix plus a dollar-neutral tilt toward 0050.

6 CORRELATION AS A COSINE

- [ ] interpret a sample correlation as the cosine of the angle between centred return vectors, and bound the standard deviation of a long-only portfolio

A return series over T dates is a vector with one entry per date. Subtracting its mean from every entry **centres** it: $\tilde{\mathbf{x}} := \mathbf{x} - \bar{x}\mathbf{1}$, with $\bar{x} = \mathbf{1}^\top\mathbf{x}/T$. With divisor T, the standard deviation is $s_x = \|\tilde{\mathbf{x}}\|/\sqrt{T}$, and the covariance of two series is $\tilde{\mathbf{x}}^\top\tilde{\mathbf{y}}/T$, the average product of deviations in LM7.

RESULT **A sample correlation is a cosine**

For two return series whose entries are not all equal, the sample correlation is

$$\rho := \frac{\tilde{\mathbf{x}}^\top\tilde{\mathbf{y}}/T}{s_x s_y} = \frac{\tilde{\mathbf{x}}^\top\tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\|\,\|\tilde{\mathbf{y}}\|} = \cos\theta, \tag{8.9}$$

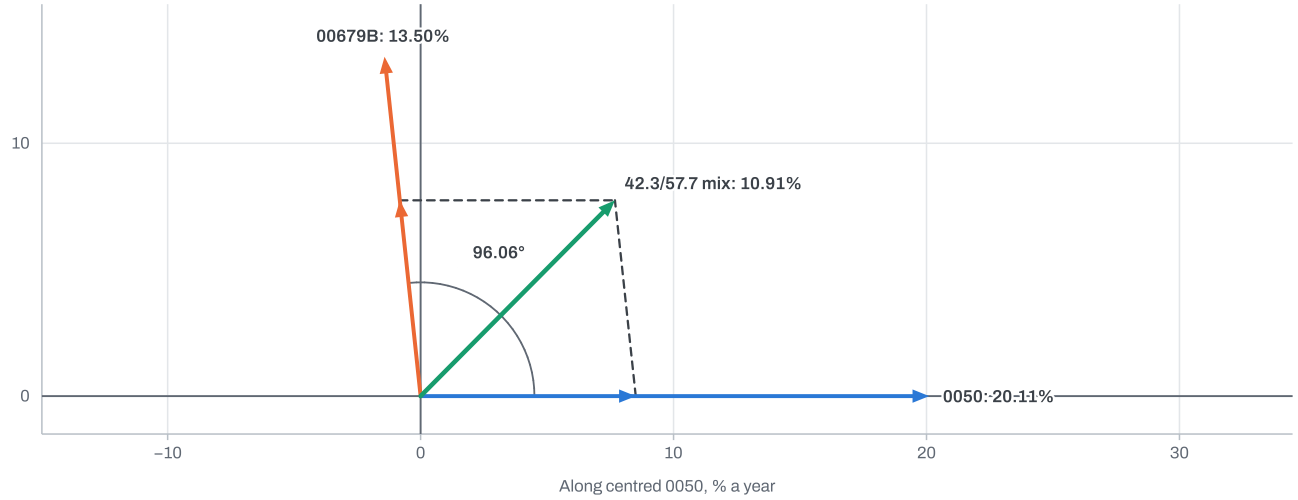
where θ is the angle between the centred vectors. Hence $-1 \le \rho \le 1$, with $\rho = \pm 1$ exactly when $\tilde{\mathbf{y}} = a\tilde{\mathbf{x}}$ for a number $a \neq 0$.

The factors T cancel in (8.9), and so would any other divisor, $T - 1$ included. Annualization and units cancel too, because multiplying a vector by a positive number does not change its direction. The bound $|\rho| \le 1$ is (8.6), with its equality case.

Two series in their own plane. Two vectors span a plane, even in 2,363 dimensions, and a drawing in that plane needs only their lengths and their angle. Exhibit 6 draws the centred 0050 and 00679B vectors of LM7, scaled so that each length is an annualized volatility $s\sqrt{A}$. The mix $0.423\,\tilde{\mathbf{x}}_{0050} + 0.577\,\tilde{\mathbf{x}}_{00679B}$ is the centred return vector of the daily-rebalanced mix, and its length is that mix's volatility.

Exhibit 6: 0050 and 00679B drawn in their own plane, daily total returns 2017–2026

Perpendicular to it, in the plane of the two vectors, % a year



Note: the centred vectors of LM7 ($T = 2{,}363$), scaled by $\sqrt{A/T}$ with $A = 243.0543$. Their lengths are 20.11% for 0050 and 13.50% for 00679B, and their angle is 96.06° ($\rho = -0.1056$). The 42.3/57.7 mix is the diagonal of the dashed parallelogram, of length 10.91%.

Standardized series. Dividing a centred vector by its standard deviation gives it length $\sqrt{T}$, and the squared distance between two such vectors depends only on ρ:

$$\left\| \frac{\tilde{\mathbf{x}}}{s_x} - \frac{\tilde{\mathbf{y}}}{s_y} \right\|^2 = T + T - 2\,\frac{\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}}}{s_x s_y} = 2T(1-\rho). \tag{8.10}$$

Per date, the root-mean-square gap between two standardized series is $\sqrt{2(1-\rho)}$: 0 for identical series, $\sqrt{2}$ for uncorrelated ones and 2 for opposite ones. LM31 builds a tree of assets from half this distance.

A bound on a portfolio's risk. A daily-rebalanced portfolio's centred return vector is $\sum_i w_i \tilde{\mathbf{x}}_i$, so its standard deviation is a length divided by $\sqrt{T}$. For non-negative weights the triangle inequality (8.8), applied one term at a time, bounds that length:

$$\begin{aligned} s_p = \frac{\left\| \sum_i w_i \tilde{\mathbf{x}}_i \right\|}{\sqrt{T}} &\le \sum_i w_i \frac{\|\tilde{\mathbf{x}}_i\|}{\sqrt{T}} \\ &= \sum_i w_i s_i \qquad (w_i \ge 0). \end{aligned} \tag{8.11}$$

Equality needs the centred vectors of all held assets to point the same way, that is, every correlation among them equal to 1. With short positions the bound becomes $\sum_i |w_i| s_i$.

EXAMPLE 6

A five-session correlation by hand

The table gives the daily returns $\mathbf{x}$ of 0050 and $\mathbf{y}$ of 00679B on 7–11 April 2025, in percent. Compute their correlation as a cosine, the angle, and the cosine of the uncentred vectors.

Date	0050, x_t	00679B, y_t
7 Apr	−10.0000	1.8072
8 Apr	−3.2513	−3.5832
9 Apr	−4.6003	−4.1937
10 Apr	9.9863	2.8826
11 Apr	2.3321	−3.5628

SOLUTION

The means are $\bar{x} = -1.10664$ and $\bar{y} = -1.32998$. Subtracting them gives the centred vectors:

Date	$\tilde{x}_t$	$\tilde{y}_t$	$\tilde{x}_t\tilde{y}_t$
7 Apr	−8.8934	3.1372	−27.9004
8 Apr	−2.1447	−2.2532	4.8324
9 Apr	−3.4937	−2.8637	10.0049
10 Apr	11.0929	4.2126	46.7300
11 Apr	3.4387	−2.2328	−7.6779
Sum			25.9890

With $\|\tilde{\mathbf{x}}\|^2 = 230.775$ and $\|\tilde{\mathbf{y}}\|^2 = 45.851$, (8.9) gives $\rho = 25.989/\sqrt{230.775 \times 45.851} = 0.2527$, an angle of 75.4°. Without centring, $\mathbf{x}^\top\mathbf{y} = 33.348$, $\|\mathbf{x}\|^2 = 236.899$ and $\|\mathbf{y}\|^2 = 54.695$, so the uncentred cosine is 0.293.

Over the full sample of LM7 the pair has $\rho = -0.1056$, an angle of 96.1°. One crash week put the funds 75° apart; a decade put them slightly more than 90° apart.

EXAMPLE 7

The bound (8.11) for the mix of LM7

Over the sample of LM7, 0050 and 00679B have annualized volatilities $s_{0050}\sqrt{A} = 20.106\%$ and $s_{00679B}\sqrt{A} = 13.496\%$ and $\rho = -0.1056$. Compute the bound (8.11) and the volatility of the mix with weight $w = 0.423$ in 0050.

SOLUTION

The bound is $0.423 \times 20.106\% + 0.577 \times 13.496\% = 16.292\%$. With $w = 0.423$, the two-asset variance of LM7, in squared percentage points, is

$$\begin{aligned} & w^2(20.106)^2 + (1-w)^2(13.496)^2 \\ & \quad + 2w(1-w)(-0.1056) \\ & \quad \times (20.106)(13.496) = 118.99, \end{aligned}$$

so the mix's volatility is $\sqrt{118.99} = 10.908\%$, two thirds of the bound. In Exhibit 6, the angle of 96° makes the diagonal much shorter than the sum of the two sides.

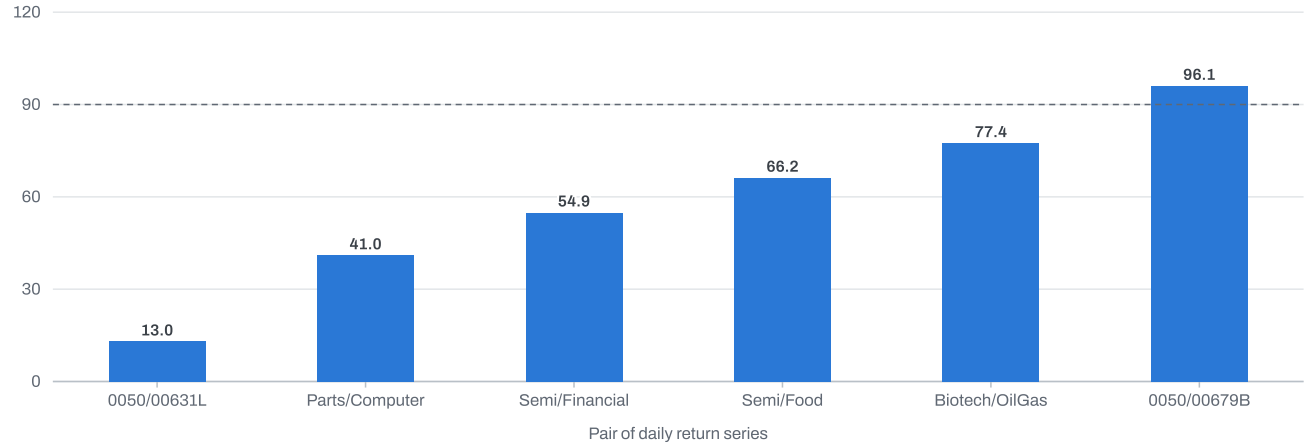
PITFALL **An uncentred cosine is not a correlation**

On long samples the daily means are tiny beside the daily moves, so skipping the centring changes little. For Semiconductor and FinancialInsurance over 2008–2026, the uncentred cosine is 0.5758 against $\rho = 0.5749$. On five sessions it shows, 0.293 against 0.253 in Example 6. Centre first, or use `np.corrcoef`, which centres for you.

The angles answer the case's third question (Exhibit 7). 0050 and 00631L, the daily 2× fund on the same index, point almost the same way, at 13.0°. The 378 pairs of the 28 sector indices lie between 41.0° and 77.4°, and the bond fund sits at 96.1° from 0050. Orthogonal centred vectors describe the sample's dates; they do not mean that two assets are unrelated (LM16).

Exhibit 7: Correlation and angle: six pairs of centred daily return vectors

Angle between the centred return vectors, degrees



Pair	Correlation ρ	Angle θ
0050 / 00631L	0.9744	13.0°
ElectronicPartsComponents / ComputerPeripheralEquipment (largest sector pair)	0.7547	41.0°
Semiconductor / FinancialInsurance	0.5749	54.9°
Semiconductor / Food	0.4035	66.2°
BiotechnologyMedicalCare / OilGasElectricity (smallest sector pair)	0.2175	77.4°
0050 / 00679B	−0.1056	96.1°

Note: the angle is $\theta = \arccos \rho$; the dashed line marks 90°, $\rho = 0$. The ETF pairs use total returns: LM1's 2,902 returns from 3 November 2014 for 00631L, LM7's 2,363 from 18 January 2017 for 00679B. The sector pairs use price returns of U28, the 28 leaf indices among the 30 TWSE sector indices that FinMind serves daily: all except the aggregates Electronic and ChemicalBiotechnologyMedicalCare. Their 378 pairs share 4,382 returns, from 2 December 2008 to 8 October 2026. Source: FinMind `TaiwanStockPrice` and `TaiwanStockDividendResult`, retrieved 10 October 2026 (ETFs) and 11 October 2026 (sector indices, with 524 copied closes on 38 dates in this sample replaced by the 13:30:00 values of `TaiwanStockEvery5SecondsIndex`); TWSE IndexS03 (the sector indices are price indices). Computations: `data/lm08_numbers.py`.

> **NOTE** **Research hygiene**
>
> Every correlation and angle in this module is **descriptive** in the sense of the Research hygiene page: a summary of the whole stated sample, not a forecast. Nothing here is fitted or tested on later data. The five-session cosine is set beside the full-sample value to show noise, not as a stability check.

導讀 **相關係數就是夾角的餘弦**

把一段報酬序列減去平均數（mean），得到中心化（centring）的向量（vector）；兩個中心化向量夾角（angle）的餘弦（cosine），正好就是樣本相關係數（sample correlation coefficient）。所以相關係數一定落在 −1 與 1 之間，除數用 T 還是 $T-1$、要不要年化，都不影響結果。0050 與 00631L 夾角 13°，幾乎同方向；0050 與 00679B 夾角 96°，略大於直角。只看五個交易日，0050 與 00679B 的夾角是 75°，那只是雜訊；完整樣本的數字才描述這段歷史。只作多的組合，標準差（standard deviation）不會超過個別標準差的加權平均，這是三角不等式（triangle inequality）的結果。

KNOWLEDGE CHECK 5

Express every 0050 return in percent (times 100) and annualize every 00679B return (times A). What happens to their correlation, and to the uncentred cosine?

ANSWER

Neither changes. A positive multiple of a vector has the same direction, and centring commutes with the scaling.

7 MATRICES AS DATA

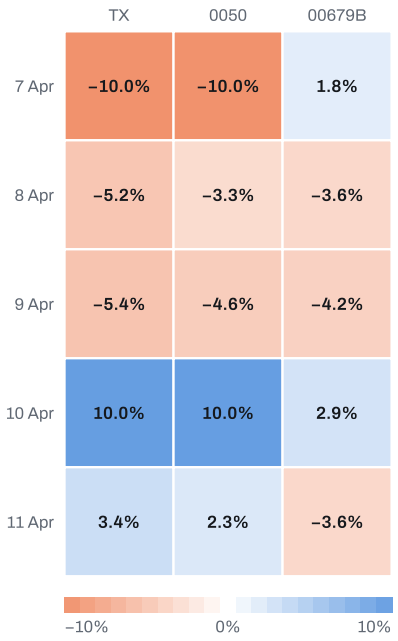
- [ ] organize returns in a data matrix and compute a daily-rebalanced portfolio's returns $X\mathbf{w}$ and a fixed book's P&L $Y\mathbf{h}$

A **matrix** is a rectangular table of numbers. Stack the daily return vectors of N assets over T dates, one row per date, and you get the **data matrix**

$$X = \begin{pmatrix} x_{11} & \cdots & x_{1N} \\ \vdots & & \vdots \\ x_{T1} & \cdots & x_{TN} \end{pmatrix} = \begin{pmatrix} \mathbf{R}_1^\top \\ \vdots \\ \mathbf{R}_T^\top \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_N \end{pmatrix}.$$

Entry x_{tj} is the return of asset j on date t: date first, asset second. Row t is the vector $\mathbf{R}_t$ written as a row, and column j is the return series $\mathbf{x}_j$ of asset j, a vector with T entries. Exhibit 8 shows the 5×3 matrix of the case week.

Exhibit 8: The week as a data matrix: daily returns of TX, 0050 and 00679B, 7–11 April 2025



Note: rows are sessions and columns instruments; each cell is a simple daily return, TX from its daily settlement price. Row 1 runs from the close of 2 April to the close of 7 April, so it spans the closed days 3–6 April. Source: FinMind `TaiwanFuturesDaily` and `TaiwanStockPrice`, retrieved 10 October 2026.

The **transpose** $X^\top$ swaps rows and columns. It is $N \times T$, and its entry in row j and column t is x_{tj}. A column vector is an $N \times 1$ matrix and its transpose a $1 \times N$ row, which is why the dot product is written $\mathbf{x}^\top \mathbf{y}$.

A portfolio's returns. Multiply X by a weight vector $\mathbf{w}$. Entry t of the product is the dot product of row t with $\mathbf{w}$, the portfolio's return on date t by (8.3). Read by columns, the same product is a linear combination of the assets' return series:

$$X\mathbf{w} = \begin{pmatrix} \mathbf{R}_1^\top \mathbf{w} \\ \vdots \\ \mathbf{R}_T^\top \mathbf{w} \end{pmatrix} = w_1 \mathbf{x}_1 + \cdots + w_N \mathbf{x}_N. \tag{8.12}$$

In the words of LM7, a matrix times a vector is the vector of each row's dot product with it. Every row uses the same $\mathbf{w}$, so $X\mathbf{w}$ is the return series of a portfolio that resets its weights to $\mathbf{w}$ at every close: it is **rebalanced daily**. LM3 and LM7 assumed daily rebalancing, and (8.12) shows where the assumption sits.

A fixed book's P&L. A book that does not trade keeps its holdings $\mathbf{h}$. Let Y be the $T \times N$ matrix of price changes, with row t equal to $\Delta\mathbf{P}_t^\top$. By (8.2), the book's daily P&Ls are

$$Y\mathbf{h} = \begin{pmatrix} \Delta\mathbf{P}_1^\top \mathbf{h} \\ \vdots \\ \Delta\mathbf{P}_T^\top \mathbf{h} \end{pmatrix} = \begin{pmatrix} \Delta W_1 \\ \vdots \\ \Delta W_T \end{pmatrix}. \tag{8.13}$$

EXAMPLE 8

A 60/40 mix over the week, rebalanced daily

Compute $X\mathbf{w}$ for $\mathbf{w} = (0,\ 0.6,\ 0.4)^{\top}$, that is 60% 0050, 40% 00679B and no TX, over 7–11 April. Compare the sum of the five returns with the return of the week.

SOLUTION

Row 1 is $0.6 \times (-10.000\%) + 0.4 \times 1.807\% = -5.277\%$, and the five rows give, in percent,

$$X\mathbf{w} = \begin{pmatrix} -5.277 \\ -3.384 \\ -4.438 \\ 7.145 \\ -0.026 \end{pmatrix}.$$

Their sum is −5.98%. The mix compounds, so its return over the week is $\prod_t (1 + R_{p,t}) - 1 = -6.32\%$.

PITFALL **A fixed book's exposures drift**

The product $Y\mathbf{h}$ keeps the holdings fixed, while $X\mathbf{f}$ would keep the exposures fixed. The two differ after the first day, because prices move the exposures. At the close of 7 April the book's equity was NT$10,057,400, W_0 plus that day's NT$57,400. Against that equity, the TX exposure had shrunk from −0.426 on 2 April to −0.381, and the 0050 exposure from 0.422 to 0.378.

On 8 April the fixed book returned −0.328% of its equity at the close of 7 April, while the exposures of 2 April give $\mathbf{f}^{\top}\mathbf{R}_{8\ \mathrm{Apr}} = -0.226\%$. Recompute exposures from the latest prices before using (8.4).

導讀 **資料矩陣：一列一天、一行一資產**

把每天的報酬向量（vector）一列一列疊起來，就是資料矩陣（data matrix）X。第 t 列（row）是第 t 天各資產的報酬，第 j 行（column）是資產 j 的整段報酬序列。矩陣乘向量有兩種讀法：按列看，每一列與權重做內積（dot product），得到當天的組合報酬；按行看，是各資產報酬序列的線性組合（linear combination）。每一天都用同一組權重，所以 $X\mathbf{w}$ 描述的是每天收盤都再平衡（rebalancing）回固定權重的組合。持股固定、不交易的帳戶要用價格變動矩陣 Y，$Y\mathbf{h}$ 是每天的損益，而它的曝險（exposure）會隨價格漂移。

KNOWLEDGE CHECK 6

Trader A rebalances to 60/40 in 0050 and 00679B at every close. Trader B buys 60/40 on 2 April and holds. Which trader's daily returns are $X\mathbf{w}$, and which daily P&Ls are $Y\mathbf{h}$ for some $\mathbf{h}$?

ANSWER

Trader A's returns are $X\mathbf{w}$, because the same weights apply every day. Trader B's holdings stay fixed, so B's P&Ls are $Y\mathbf{h}$ with $\mathbf{h}$ the shares bought on 2 April. B's weights drift after the first day.

8 MATRIX MULTIPLICATION, TRANSPOSE AND IDENTITY

- ☐ multiply and transpose matrices, check their shapes, and apply associativity, non-commutativity and $(MK)^\top = K^\top M^\top$
- ☐ implement these operations in NumPy and Polars, with dates aligned by a join and assets in a fixed order

Equation (8.12) multiplies a matrix by one vector. Multiplying by several vectors at once, stacked as the columns of a matrix, gives the general product.

DEFINITION **Matrix product**

If M is $n \times N$ and K is $N \times m$, their **product** MK is the $n \times m$ matrix with entries

$$(MK)_{ij} := \sum_{\ell=1}^{N} m_{i\ell}\, k_{\ell j}, \tag{8.14}$$

the dot product of row i of M with column j of K. The inner sizes must agree; otherwise the product is not defined.

Put the weights of m portfolios side by side as the columns of an $N \times m$ matrix K. Column j of XK is X times column j of K, so each column of XK is one portfolio's daily return series. Exhibit 9 does this for three portfolios in the case week, with rows in the order (TX, 0050, 00679B):

$$K = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0.6 & 0.5 \\ 0 & 0.4 & 0.5 \end{pmatrix}.$$

Exhibit 9: Three portfolios at once: the columns of XK, 7–11 April 2025

Session	100% 0050	60/40	50/50
7 Apr	−10.000%	−5.277%	−4.096%
8 Apr	−3.251%	−3.384%	−3.417%
9 Apr	−4.600%	−4.438%	−4.397%
10 Apr	+9.986%	+7.145%	+6.434%
11 Apr	+2.332%	−0.026%	−0.615%
Sum of daily returns	−5.533%	−5.980%	−6.091%
Compounded week	−6.506%	−6.319%	−6.328%

Note: portfolios rebalanced at every close; 60/40 and 50/50 are weights in 0050 and 00679B. The last row compounds the unrounded daily returns, so compounding the rounded entries can change its last digit. Computations: `data/lm08_numbers.py`.

Rules of the product. The product is **associative**: $(MK)\mathbf{x} = M(K\mathbf{x})$. By (8.14), entry i of $(MK)\mathbf{x}$ is

$$\begin{aligned}\sum_j \Big(\sum_\ell m_{i\ell}k_{\ell j}\Big)x_j &= \sum_\ell m_{i\ell}\Big(\sum_j k_{\ell j}x_j\Big)\\ &= \sum_\ell m_{i\ell}(K\mathbf{x})_\ell,\end{aligned}$$

which is entry i of $M(K\mathbf{x})$. Applied to each column of a third matrix, the same identity makes a product of three matrices associative. Holding the three portfolios of Exhibit 9 in proportions $\mathbf{x}$ therefore gives the returns of the combined weights $K\mathbf{x}$.

The product is **not commutative**. The product KM may not exist, and when both products exist they usually differ:

$$\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}1&0\\1&1\end{pmatrix} = \begin{pmatrix}2&1\\1&1\end{pmatrix},$$
$$\begin{pmatrix}1&0\\1&1\end{pmatrix}\begin{pmatrix}1&1\\0&1\end{pmatrix} = \begin{pmatrix}1&1\\1&2\end{pmatrix}.$$

PROPOSITION **Transpose of a product**

For an $n \times N$ matrix M and an $N \times m$ matrix K,

$$(MK)^\top = K^\top M^\top. \tag{8.15}$$

Proof. The entry in row i and column j of $(MK)^\top$ is entry (j,i) of MK. By (8.14) it equals

$$\sum_{\ell} m_{j\ell} k_{\ell i} = \sum_{\ell} (K^{\top})_{i\ell} (M^{\top})_{\ell j},$$

which is entry (i, j) of $K^{\top} M^{\top}$. ■

For a vector, (8.15) gives $(X\mathbf{w})^{\top} = \mathbf{w}^{\top} X^{\top}$, a row.

Identity. The $N \times N$ **identity matrix** I has ones on its diagonal and zeros elsewhere, so $MI = M$ and $IK = K$. It plays the role of 1. Dividing by a matrix means multiplying by its inverse, which exists only sometimes (LM9). The dot products of all centred columns form a square table $\tilde{X}^{\top}\tilde{X}$; LM10 divides it by T and calls it the sample covariance matrix.

Summing over dates. Multiplying a column by $\mathbf{1}^{\top}$ adds its entries. For a fixed book the sum telescopes, because the daily price changes sum to the change over the period:

$$\mathbf{1}^{\top} Y \mathbf{h} = \sum_{t=1}^{T} \Delta \mathbf{P}_t^{\top} \mathbf{h} = (\mathbf{P}_T - \mathbf{P}_0)^{\top} \mathbf{h}.$$

This sum is the book's exact P&L over the period. For a daily-rebalanced portfolio, $\mathbf{1}^{\top} X \mathbf{w}$ is only the sum of the daily returns, and that sum is no period return, because returns compound.

EXAMPLE 9

Why the hedged book lost NT$124,200

Compute $Y\mathbf{h}$ for the case book over 7–11 April and its sum, and split the week's loss by line. Then rank the portfolios of Exhibit 9 by their summed and by their compounded returns.

SOLUTION

By (8.13), the daily P&Ls in NT$ are

$$Y\mathbf{h} = \begin{pmatrix} 57{,}400 \\ -33{,}000 \\ -95{,}200 \\ 87{,}800 \\ -141{,}200 \end{pmatrix},$$

so $\mathbf{1}^{\top} Y\mathbf{h} = -124{,}200$. The telescoping sum gives the same figure line by line:

$$\begin{aligned} &(-1)(200)(19{,}543 - 21{,}296) \\ &\quad + 24{,}000\,(164.55 - 176.00) \\ &\quad + 100{,}000\,(27.88 - 29.88) \\ &= 350{,}600 - 274{,}800 - 200{,}000. \end{aligned}$$

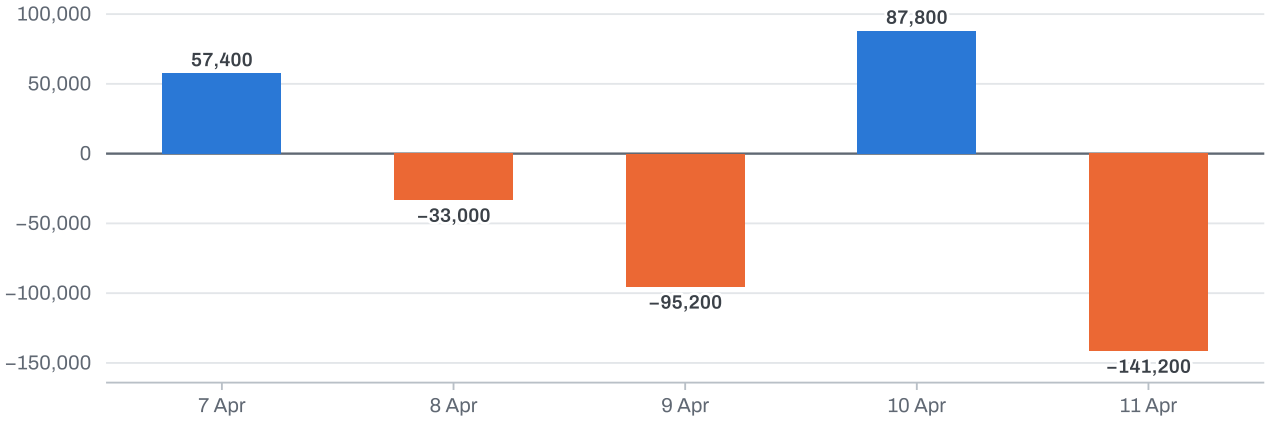
The TX and 0050 lines together made NT$75,800, because TX fell 8.2% over the week and 0050 only 6.5%. The loss came from the bond: 00679B fell from 29.88 to 27.88, a loss of NT$200,000 on a position that nothing hedged.

Summed returns rank 100% 0050 first, but compounded returns rank it last, at −6.506%. Its swings of −10% and +10% cost more than the sum shows.

A fixed book over 7–11 April 2025: the daily P&L Y h

INTERACTIVE

Daily P&L of the book, NT$



P&L on 7 April

NT$+57,400

Week, the sum of the bars

NT$−124,200

Net exposure, 2 April

NT$2,952,800

0.295 × equity

Gross exposure, 2 April

NT$11,471,200

1.147 × equity

The holdings stay fixed all week, so each bar is one entry of Y h: the session's price changes times the holdings, summed over the three lines. Bars above zero are gains and bars below zero losses. TX moves NT$200 a point per contract. Exposures are position values at the close of 2 April, also shown as a multiple of equity of NT$10 million.

IN PYTHON Shapes and products in NumPy

`@` is the matrix product and `.T` the transpose. NumPy treats a 1-D array as a row or a column, whichever the product needs, so print shapes when they matter and use `reshape(-1, 1)` for an explicit column.

PYTHON

```python
import numpy as np

# TX settlement (points), 0050 and 00679B closes, 2 and 7-11 April 2025
P = np.array([[21296, 176.00, 29.88], [19167, 158.40, 30.42],
              [18169, 153.25, 29.33], [17184, 146.20, 28.10],
              [18902, 160.80, 28.91], [19543, 164.55, 27.88]])
X = P[1:] / P[:-1] - 1                      # 5 x 3 simple returns
K = np.array([[0, 0, 0], [1, 0.6, 0.5], [0, 0.4, 0.5]])
print(X.shape, K.shape, (X @ K).shape)      # (5, 3) (3, 3) (5, 3)
w = K[:, 1]                                 # 60/40 weights, a 1-D array
print(w.shape, w.T.shape)                   # (3,) (3,): no orientation
print(np.allclose((X @ K).T, K.T @ X.T))    # True: (MK)^T = K^T M^T
h = np.array([-1, 24_000, 100_000])
Y = np.diff(P * [200, 1, 1], axis=0)        # price changes, NT$ per unit
print((Y @ h).round(), (Y @ h).sum().round())
# [  57400.  -33000.  -95200.   87800. -141200.] -124200.0
```

IN PYTHON Companion lab

The Polars-first notebook `LM08_lab.ipynb` joins the three instruments on their dates and rebuilds the ETF pairs and the sector sample. It reproduces the exhibits and the computational examples, and solves the computational practice problems.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

The second trader of Section 3 held $(-1,\ 20{,}000,\ 100{,}000)^{\top}$ all week. Use the telescoping sum to find the book's P&L over 7–11 April.

ANSWER

The weekly price changes are $-1{,}753 \times 200 = -350{,}600$ per TX contract, -11.45 per 0050 share and -2.00 per 00679B share. So

$$\begin{aligned}&\mathbf{h}^{\top}(\mathbf{P}_{11\ \text{Apr}} - \mathbf{P}_{2\ \text{Apr}})\\ &= 350{,}600 - 229{,}000 - 200{,}000\\ &= -78{,}400\ \text{NT\$}.\end{aligned}$$

SUMMARY

- A vector lists numbers in a fixed order, each with a unit and an index of asset and date; vectors add only when all three agree.
- The dot product (8.1) gives P&L, $\mathbf{h}^{\top}\Delta\mathbf{P}$ (8.2), and portfolio return, $\mathbf{w}^{\top}\mathbf{R}$ (8.3), for simple returns only. On 7 April 2025 the case book made NT\$57,400, all but NT\$3,400 of it on the bond line.
- The P&L is exactly $W_0\mathbf{f}^{\top}\mathbf{R}$ (8.4). Net exposure 0.295 and gross exposure 1.147 describe the same book.
- The Cauchy–Schwarz inequality (8.6), with equality only for parallel vectors, defines angles (8.7) and gives the triangle inequality (8.8).
- A sample correlation is the cosine of the angle between centred vectors (8.9): 13.0° for 0050 and 00631L, 96.1° for 0050 and 00679B, 41.0° to 77.4° across the sector pairs.
- For long-only weights $s_p \le \sum_i w_i s_i$ (8.11): 10.91% against 16.29% for the mix of LM7.
- In a data matrix, $X\mathbf{w}$ (8.12) is a daily-rebalanced portfolio's return series and $Y\mathbf{h}$ (8.13) a fixed book's P&L. The case book lost NT\$124,200 over the week, and its unhedged bond line NT\$200,000.
- Matrix products (8.14) are associative but not commutative, and $(MK)^{\top} = K^{\top}M^{\top}$ (8.15). Summed daily returns are no period return.

PRACTICE PROBLEMS

1. Which expression gives the one-day return of a fully invested portfolio with weights $\mathbf{w}$?
   - **A.** The weighted sum of log returns, $\mathbf{w}^{\top}\mathbf{r}$
   - **B.** The weighted sum of simple returns, $\mathbf{w}^{\top}\mathbf{R}$
   - **C.** The holdings times the simple returns, $\mathbf{h}^{\top}\mathbf{R}$

2. A book holds two TX April contracts long, no 0050 and 200,000 shares of 00679B, so $\mathbf{h} = (2,\ 0,\ 200{,}000)^\top$. On 7 April TX settled at 19,167 and 00679B closed at 30.42; on 8 April they were 18,169 and 29.33. Compute the book's P&L on 8 April.

3. Take the book of Problem 2 at the close of 2 April, with equity of NT$10 million. Compute its exposures $\mathbf{f}$ from Exhibit 1, its net and gross exposure, and check (8.4) for 7 April.

4. Two weight vectors are $(0.5,\ 0.3,\ 0.2)^\top$ and $(0.4,\ 0.3,\ 0.3)^\top$. Compute their Euclidean norms, the distance between them, the distance relative to the norm of the first, and the 1-norm of their difference. What does the 1-norm measure for the trade from the first mix to the second?

5. Prove that $\|\mathbf{x}+\mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\|$ holds exactly when one of the vectors is a non-negative multiple of the other.

6. For $\mathbf{x} = (1,\ 2,\ 2)^\top$ and $\mathbf{y} = (2,\ 1,\ 2)^\top$, find the angle between the vectors and the number a for which $\mathbf{x} - a\mathbf{y}$ is orthogonal to $\mathbf{y}$.

7. Four daily returns, in percent, are $\mathbf{x} = (2,\ 4,\ 0,\ 2)^\top$ and $\mathbf{y} = (1,\ 3,\ 1,\ 3)^\top$. Compute their correlation as the cosine of the centred vectors. Then compute it from the covariance and standard deviations with divisor $T = 4$, and again with divisor $T - 1 = 3$.

8. The daily returns of 0050 and 00679B have $\rho = -0.106$. The angle between their centred vectors is *closest* to:

   A. 84°

   B. 90°

   C. 96°

9. Among the 378 pairs of the 28 sector indices, the largest correlation is 0.7547 (ElectronicPartsComponents and ComputerPeripheralEquipment) and the smallest 0.2175 (BiotechnologyMedicalCare and OilGasElectricity). For each pair, find the angle between the centred vectors and the root-mean-square gap per date between the standardized series.

10. Over 7–11 April 2025, which of these is exact?

    A. The sum of a fixed book's daily P&Ls, $\mathbf{1}^\top Y\mathbf{h}$, taken as the book's P&L for the week

    B. The sum of a portfolio's daily returns, $\mathbf{1}^\top X\mathbf{w}$, taken as the week's return

    C. The daily P&Ls $W_0 X\mathbf{f}$, with the exposures of 2 April, taken as the fixed book's P&L on each day of the week

11. *(Python)* With the lab's data, keep the TX rows with `contract_date` 202504 and use their `settlement_price`. Join them with the closes of 0050 and 00679B on their common dates in April 2025, and build X for 7–11 April. Compute $X\mathbf{w}$ for the 60/40 mix of Example 8, and compare the cosine of the centred 0050 and 00679B columns with `np.corrcoef`.

SOLUTIONS

1. **B** is correct. Simple returns combine with weights, (8.3). A fails because log returns do not combine with weights: the portfolio's log return is $\ln(1 + \mathbf{w}^\top \mathbf{R})$. C multiplies contracts and shares by pure numbers, so $\mathbf{h}^\top \mathbf{R}$ is neither a return nor a P&L; the P&L is $\mathbf{h}^\top \Delta \mathbf{P}$ (8.2).

2. The lines are $2 \times 200 \times (18{,}169 - 19{,}167) = -399{,}200$ and $200{,}000 \times (29.33 - 30.42) = -218{,}000$, so the book lost NT$617,200 on 8 April.

3. The position values on 2 April are $2 \times 200 \times 21{,}296 = 8{,}518{,}400$, 0 and $200{,}000 \times 29.88 = 5{,}976{,}000$, so $\mathbf{f} = (0.85184,\ 0,\ 0.5976)^\top$. Net and gross exposure are both 1.44944, because every position is long. With the returns of Exhibit 1, $\mathbf{f}^\top \mathbf{R} = -0.085160 + 0.010800 = -0.074360$, and $W_0\, \mathbf{f}^\top \mathbf{R} = -743{,}600$. Directly, $2 \times (-425{,}800) + 200{,}000 \times 0.54 = -743{,}600$ NT$.

4. The norms are $\sqrt{0.38} = 0.6164$ and $\sqrt{0.34} = 0.5831$. The difference is $(-0.1,\ 0,\ 0.1)^\top$, so the distance is 0.1414, and relative to the first norm it is $0.1414/0.6164 = 0.2294$. The 1-norm of the difference, 0.2, measures the trade's turnover. The trade sells 10% of capital in one asset and buys 10% in another, a two-way turnover of 20% (10% one way).

5. By the derivation of (8.8), equality needs $\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\|$. If $\mathbf{y} = \mathbf{0}$, it holds, and $\mathbf{y} = 0 \cdot \mathbf{x}$. Otherwise equality in (8.6) makes $\mathbf{x} = a\mathbf{y}$ for a number a, and then $\mathbf{x}^\top \mathbf{y} = a\|\mathbf{y}\|^2$ equals $\|\mathbf{x}\|\|\mathbf{y}\| = |a|\|\mathbf{y}\|^2$ only when $a \geq 0$. Conversely, if $\mathbf{x} = a\mathbf{y}$ with $a \geq 0$, both sides equal $(1 + a)\|\mathbf{y}\|$.

6. The dot product is $2 + 2 + 4 = 8$ and both lengths are 3, so $\cos\theta = 8/9$ and $\theta = 27.27°$. The proof of (8.6) gives $a = \mathbf{x}^\top \mathbf{y}/\|\mathbf{y}\|^2 = 8/9$. Check: $\mathbf{x} - \frac{8}{9}\mathbf{y} = (-\frac{7}{9},\ \frac{10}{9},\ \frac{2}{9})^\top$, whose dot product with $\mathbf{y}$ is $(-14 + 10 + 4)/9 = 0$.

7. Both means are 2, so $\tilde{\mathbf{x}} = (0,\ 2,\ -2,\ 0)^\top$ and $\tilde{\mathbf{y}} = (-1,\ 1,\ -1,\ 1)^\top$. Then $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} = 4$, $\|\tilde{\mathbf{x}}\| = \sqrt{8}$ and $\|\tilde{\mathbf{y}}\| = 2$, so $\rho = 4/(2\sqrt{8}) = 0.7071$, an angle of 45°. With divisor 4, the covariance is 1 and the standard deviations are $\sqrt{2}$ and 1, so $\rho = 1/\sqrt{2}$. With divisor 3, they are $4/3$, $\sqrt{8/3}$ and $\sqrt{4/3}$, and $\rho = (4/3)/\sqrt{32/9} = 1/\sqrt{2}$ again.

8. **C** is correct. The angle is $\arccos(-0.106) = 96.1°$, a little more than a right angle. A, 83.9°, drops the sign of ρ. B treats a small correlation as zero.

9. By (8.10), the root-mean-square gap per date between the standardized series is $\sqrt{2(1 - \rho)}$. For 0.7547, $\theta = \arccos 0.7547 = 41.00°$ and $\sqrt{2(1 - 0.7547)} = 0.7004$. For 0.2175, $\theta = 77.44°$ and $\sqrt{2(1 - 0.2175)} = 1.2510$. The root-mean-square daily gap is 0.70 standard deviations for the two electronics sectors and 1.25 for the least related pair.

10. **A** is correct. The daily P&Ls of a fixed book telescope to $\mathbf{h}^\top(\mathbf{P}_{11\ \text{Apr}} - \mathbf{P}_{2\ \text{Apr}})$. B ignores compounding: for the 60/40 mix the sum is −5.98% while the week's return is −6.32%. C ignores the drift of the exposures, so it is exact only on 7 April. On 8 April it gives a P&L of −22,628 against the book's −33,000, a miss of NT$10,372.

11. The lab keeps contract 202504, joins the three files on `date` with an inner join and keeps 2 to 11 April; its $X\mathbf{w}$ matches Example 8. With its frame `wk` of the five returns:

PYTHON

```python
x, y = wk["R_0050"].to_numpy(), wk["R_00679B"].to_numpy()
xc, yc = x - x.mean(), y - y.mean()
print(xc @ yc / np.linalg.norm(xc) / np.linalg.norm(yc))   # 0.2526...
print(np.corrcoef(x, y)[0, 1])                             # the same
```

The two numbers agree, at 0.2526 to four decimals. Example 6 got 0.2527 because it worked from its table's returns, in percent rounded to four decimals.

GLOSSARY

Term	中文	Meaning
Vector	向量	Ordered list of numbers, a column here, with a unit per entry and an index of assets and dates.
Linear combination	線性組合	Sum of vectors times numbers, such as $\sum_j w_j \mathbf{x}_j$.
Dot product	內積	$\mathbf{x}^\top \mathbf{y} = \sum_k x_k y_k$, (8.1).
Exposure	曝險	Position value ÷ equity, $f_i = h_i P_{i,0} / W_0$.
Net exposure	淨曝險	$\mathbf{1}^\top \mathbf{f}$: longs minus shorts.
Gross exposure	總曝險	$\|\mathbf{f}\|_1$: longs plus shorts.
Euclidean norm	歐氏範數	Length $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$, (8.5).
Unit vector	單位向量	Vector of length 1.
Cauchy–Schwarz inequality	Cauchy–Schwarz 不等式	$\|\mathbf{x}^\top \mathbf{y}\| \le \|\mathbf{x}\| \|\mathbf{y}\|$, (8.6).
Angle	夾角	$\cos\theta = \mathbf{x}^\top \mathbf{y} / (\|\mathbf{x}\| \|\mathbf{y}\|)$, (8.7).
Orthogonal	正交	With dot product zero.
Triangle inequality	三角不等式	$\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$, (8.8).
Centred vector	中心化向量	A series minus its mean, $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$.
Correlation	相關係數	Cosine of the angle between centred vectors, (8.9).
Matrix	矩陣	Rectangular table of numbers.
Data matrix	資料矩陣	$T \times N$ matrix of returns, a row per date and a column per asset.
Transpose	轉置	Matrix with rows and columns swapped.
Daily rebalancing	每日再平衡	Resetting the weights at every close; its returns are $X\mathbf{w}$, (8.12).
Matrix product	矩陣乘積	$(MK)_{ij} = \sum_\ell m_{i\ell} k_{\ell j}$, (8.14).
Identity matrix	單位矩陣	Ones on the diagonal, zeros elsewhere; $MI = M$.

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